

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1248

Beaver Valley 138 kV
August 31, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1248. The Interconnection Customer proposes a 85.0 MW battery energy storage facility interconnecting at the Beaver Valley 138 kV substation in the Pennsylvania area. The request was received on July 13, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 456 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,258,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Septa to Newbury 138.0 kV line
Contingency: Aes Beaver Vly Partners Beaver Vly to Lighthouse Hill 345.0 kV
Loading: 157.9% (Rating: 188 MVA)
DFAX: 6.1%
Pre-project loading: 155.1%
2. Facility: Midland 230-138 kV Auto
Contingency: Midland 345 kV/230 kV Transformer Bank
Loading: 109.2% (Rating: 105 MVA)
DFAX: 12.5%
Pre-project loading: 99.2%
3. Facility: Talenenergy Martins Crk Llc Harrisburg 500/345kV XFMR
Contingency: Westwood Generation Llc to Coronet 500.0 kV
Loading: 100.2% (Rating: 1,584 MVA)
DFAX: 5.7%
Pre-project loading: 99.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Septa to Newbury 138.0 kV	Rebuild line with higher-capacity conductor	\$5,495,000	10% (\$522,000)
Midland 230/138 kV Autotransformer	Replace transformer with higher-rated unit	\$6,490,000	100% (\$6,490,000)
Talenenergy Martins Crk Llc Harrisburg 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$27,631,000	1% (\$246,000)
	Total allocated to project		\$7,258,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.